Advanced Polynomial Factoring

Concept

Use structure to factor: common factors, trinomials, grouping and square differences.

GCF first

Remove a common factor before choosing another pattern.

Split the middle

For $ax^2 + bx + c$, find product ac and sum b.

Patterns

$A^2 - B^2 = (A - B)(A + B)$; $A^2 \pm 2AB + B^2 = (A \pm B)^2$.

Grouping

Factor a shared binomial; substitute u for repeated expressions if useful.

Example 1

Factor $6x^2 + 11x + 3$.

Solution

1. $ac = 18$; choose 9 and 2 because $9 + 2 = 11$.
2. $6x^2 + 9x + 2x + 3 = 3x(2x + 3) + (2x + 3)$.
3. Result: $(3x + 1)(2x + 3)$.

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Example 2

Factor $12x^3 - 75x$ completely.

Solution

1. GCF: $3x(4x^2 - 25)$.

2. Difference of squares: $3x(2x - 5)(2x + 5)$.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Factor $3x^2 + 10x + 8$.

2. Factor $2x^2 - 7x + 3$.

3. Factor $18x^2 - 8y^2$.

4. Factor $x^2 - 6x + 9$.

5. Factor $5x(x - 2) + 3(x - 2)$.

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Practice answers | check after trying

1. $(3x + 4)(x + 2)$.

2. $(2x - 1)(x - 3)$.

3. $2(9x^2 - 4y^2) = 2(3x - 2y)(3x + 2y)$.

4. $(x - 3)^2$.

5. $(x - 2)(5x + 3)$. Expand each product to verify all coefficients.